

The routine *YOUFIT* in *LINTEK* can be used to illustrate graphically the fitting of data points by linear, quadratic, or exponential functions. In Exercises 27–31, use *YOUFIT* to try visually to fit the given data with the indicated type of graph. When this is done, enter the zero data suggested to see the computer's fit. Run twice more with the same data points but without trying to fit the data visually, and determine whether the data are best fitted by a linear, quadratic, or (logarithmically fitted) exponential function, by comparing the least-squares sums for the three cases.

27. Fit (1, 2), (4, 6), (7, 10), (10, 14), (14, 19) by a linear function.
28. Fit (2, 2), (6, 10), (10, 12), (16, 2) by a quadratic function.
29. Fit (1, 1), (10, 8), (14, 12), (16, 20) by an exponential function. Try to achieve a lower squares sum than the computer obtains with its least-squares fit that uses logarithms of y -values.
30. Repeat Exercise 29 with data (1, 9), (5, 1), (6, .5), (9, .01).
31. Fit (2, 9), (4, 6), (7, 1), (8, .1) by a linear function.

The routine *QRFAC* in *LINTEK* has an option to use a *QR*-factorization of A to find the least-squares solution of a linear system $Ax = b$,

executing back substitution on $Rx = Q^T b$ as in Eq. (12).

Recall that in *MATLAB*, if A is $n \times k$, then Q is $n \times n$ and R is $n \times k$. Cutting Q and R down to the text sizes $n \times k$ and $k \times k$, respectively, we can use the command lines

```
[Q R] = qr(A); [n, k] = size(A);
rref([R(1:k,1:k) Q(:,1:k)'*b])
```

to compute the solution of $Rx = Q^T b$.

Use *LINTEK* or *MATLAB* in this fashion for Exercises 32–37. You must supply the matrix A and the vector b .

32. Find the least-squares linear fit for the data points $(-3, 10)$, $(-2, 8)$, $(-1, 7)$, $(0, 6)$, $(1, 4)$, $(2, 5)$, $(3, 6)$.
33. Find the least-squares quadratic fit for the data points in Exercise 32.
34. Find the least-squares cubic fit for the data points in Exercise 32.
35. Find the least-squares quartic fit for the data points in Exercise 32.
36. Find the quadratic polynomial function whose graph passes through the points $(1, 4)$, $(2, 15)$, $(3, 32)$.
37. Find the cubic polynomial function whose graph passes through the points $(-1, 13)$, $(0, -5)$, $(2, 7)$, $(3, 25)$.

CHANGE OF BASIS

Recall from Section 3.4 that a linear transformation is a mapping of one vector space into another that preserves addition and scalar multiplication. Two vector spaces V and V' are isomorphic if there exists a linear transformation of V onto V' that is one-to-one. Of particular importance are the *coordinatization isomorphisms* of an n -dimensional vector space V with $\mathbb{R}^n$. One chooses an ordered basis B for V and defines $T: V \rightarrow \mathbb{R}^n$ by taking $T(\mathbf{v}) = \mathbf{v}_B$, the coordinate vector of $\mathbf{v}$ relative to B , as described in Section 3.3. Such an isomorphism describes V and $\mathbb{R}^n$ as being virtually the same as vector spaces. Thus much of the work in finite-dimensional vector spaces can be reduced to computations in $\mathbb{R}^n$. We take full advantage of this feature as we continue our study of linear transformations in this chapter.

We have seen that bases other than the standard basis E of $\mathbb{R}^n$ can be useful. For example, suppose A is an $n \times n$ matrix. Changing from the standard basis of $\mathbb{R}^n$ to a basis of eigenvectors (when this is possible) facilitates computation of the powers A^k of A using the more easily computed powers of a diagonal matrix, as described in Section 5.2. Our work in the preceding chapter gives another illustration of a desirable change of basis: recall the convenience that an orthonormal basis can provide. We expect that changing to a new basis will change coordinate vectors and matrix representations of linear transformations.

In Section 7.1 we review the notion of the coordinate vector $\mathbf{v}_B$ relative to an ordered basis B , and consider the effect of changing the ordered basis from B to B' . With this backdrop we turn to matrix representations of a linear transformation in Section 7.2, examining the relationship between matrix representations of a linear transformation $T: V \rightarrow V'$ relative to different ordered bases of V and of V' .

7.1

COORDINATIZATION AND CHANGE OF BASIS

Let $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ be an ordered basis for a vector space V . Recall that if $\mathbf{v}$ is a vector in V and $\mathbf{v} = r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \dots + r_n\mathbf{b}_n$, then the *coordinate vector of $\mathbf{v}$ relative to B* is

$$\mathbf{v}_B = [r_1, r_2, \dots, r_n].$$

For example, the coordinate vector of $\cos^2 x$ relative to the ordered basis $(1, \sin^2 x)$ in the vector space $\text{sp}(1, \sin^2 x)$ is $[1, -1]$. If this ordered basis is changed to some other ordered basis, the coordinate vector may change. In this section we consider the relationship between two coordinate vectors $\mathbf{v}_B$ and $\mathbf{v}_{B'}$ of the same vector $\mathbf{v}$.

We begin our discussion with $\mathbb{R}^n$, which we will view as a space of column vectors. Let $\mathbf{v}$ be a vector in $\mathbb{R}^n$ with coordinate vector

$$\mathbf{v}_B = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}$$

relative to an ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$, so that

$$r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \dots + r_n\mathbf{b}_n = \mathbf{v}. \quad (1)$$

Let M_B be the matrix having the vectors in the ordered basis B as column vectors; this is the **basis matrix** for B , which we display as

$$M_B = \begin{bmatrix} | & | & & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & & | \end{bmatrix}. \quad (2)$$

Equation (1), which expresses $\mathbf{v}$ as a vector in the column space of the matrix M_B , can be written in the form

$$M_B \mathbf{v}_B = \mathbf{v}. \quad (3)$$

If B' is another ordered basis for $\mathbb{R}^n$, we can similarly obtain

$$M_{B'} \mathbf{v}_{B'} = \mathbf{v}. \quad (4)$$

Equations (3) and (4) together yield

$$M_{B'} \mathbf{v}_{B'} = M_B \mathbf{v}_B. \quad (5)$$

Equation (5) is easy to remember, and it turns out to be very useful. Both M_B and $M_{B'}$ are invertible, because they are square matrices whose column vectors are independent. Thus, Eq. (5) yields

$$\mathbf{v}_{B'} = M_{B'}^{-1} M_B \mathbf{v}_B. \quad (6)$$

Equation (6) shows that, given any two ordered bases B and B' of $\mathbb{R}^n$, there exists an invertible matrix C —namely,

$$C = M_{B'}^{-1} M_B, \quad (7)$$

such that, for all $\mathbf{v}$ in $\mathbb{R}^n$,

$$\mathbf{v}_{B'} = C \mathbf{v}_B. \quad (8)$$

If we know this matrix C , we can conveniently convert coordinates relative to B into coordinates relative to B' . The matrix C in Eq. (7) is the unique matrix satisfying Eq. (8). This can be seen by assuming that D is also a matrix satisfying Eq. (8), so that $\mathbf{v}_{B'} = D \mathbf{v}_B$; then $C \mathbf{v}_B = D \mathbf{v}_B$ for all vectors $\mathbf{v}_B$ in $\mathbb{R}^n$. Exercise 41 in Section 1.3 shows that we must have $C = D$.

The matrix C in Eq. (7) and Eq. (8) is computed in terms of the ordered bases B and B' , and we will now change to the subscripted notation $C_{B,B'}$ to suggest this dependency. Thus Eq. (8) becomes

$$\mathbf{v}_{B'} = C_{B,B'} \mathbf{v}_B,$$

and the subscripts on C , read from left to right, indicate that we are changing from coordinates relative to B to coordinates relative to B' .

Because every n -dimensional vector space is isomorphic to $\mathbb{R}^n$, all our results here are valid for coordinates with respect to ordered bases B and B' in any finite-dimensional vector space. We phrase the definition that follows in these terms.

DEFINITION 7.1 Change-of-Coordinates Matrix

Let B and B' be ordered bases for a finite-dimensional vector space V . The **change-of-coordinates matrix** from B to B' is the unique matrix $C_{B,B'}$ such that $C_{B,B'} \mathbf{v}_B = \mathbf{v}_{B'}$ for all vectors $\mathbf{v}$ in V .

The term *transition matrix* is used in some texts in place of *change-of-coordinates matrix*. But we used *transition matrix* with reference to Markov chains in Section 1.7, so we avoid duplicate terminology here by using the more descriptive term *change-of-coordinates matrix*.

Equation (8), written in the form $C^{-1} \mathbf{v}_{B'} = \mathbf{v}_B$, shows that the inverse of the change-of-coordinates matrix from B to B' is the change-of-coordinates matrix from B' to B —that is, $C_{B',B} = C_{B,B'}^{-1}$.

Equation (7) shows us exactly what the change-of-coordinates matrix must be for any two ordered bases B and B' in $\mathbb{R}^n$. A direct way to compute this product $M_{B'}^{-1}M_B$, if $M_{B'}^{-1}$ is not already available, is to form the partitioned matrix

$$[M_{B'} \mid M_B]$$

and reduce it (by using elementary row operations) to the form $[I \mid C]$. We can regard this reduction as solving n linear systems, one for each column vector of M_B and all having the same coefficient matrix $M_{B'}$. From this perspective, the matrix $M_{B'}$ times the j th column vector of C (that is, the j th “solution vector”) must yield the j th column vector of M_B . This shows that $M_{B'}C = M_B$. Consequently, we must have $C = M_{B'}^{-1}M_B = C_{B,B'}$. We now have a convenient procedure for finding a change-of-coordinates matrix.

Finding the Change-of-Coordinates Matrix from B to B' in $\mathbb{R}^n$

Let $B = (b_1, b_2, \dots, b_n)$ and $B' = (b'_1, b'_2, \dots, b'_n)$ be ordered bases of $\mathbb{R}^n$. The change-of-coordinates matrix from B to B' is the matrix $C_{B,B'}$ obtained by the row reduction

$$\left[\begin{array}{ccc|ccc} b_1 & b_2 & \dots & b_n & b'_1 & b'_2 & \dots & b'_n \end{array} \right] \sim [I \mid C_{B,B'}].$$

New basis matrix Old basis matrix

EXAMPLE 1 Let $B = ([1, 1, 0], [2, 0, 1], [1, -1, 0])$ and let $E = (e_1, e_2, e_3)$ be the standard ordered basis of $\mathbb{R}^3$. Find the change-of-coordinates matrix $C_{E,B}$ from E to B , and use it to find the coordinate vector v_b of

$$v = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

relative to B .

SOLUTION Following the boxed procedure, we place the “new” basis matrix to the left and the “old” basis matrix to the right in an augmented matrix, and then we proceed with the row reduction:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & -2 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right]$$

M_B M_E

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & -1 & -\frac{1}{2} & \frac{1}{2} & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & -1 \end{array} \right]$$

Thus the change-of-coordinates matrix from E to B is

$$C_{E,B} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -1 \\ 0 & 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} & -1 \end{bmatrix}.$$

The coordinate vector of $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ relative to B is

$$\mathbf{v}_B = C_{E,B}\mathbf{v} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -1 \\ 0 & 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -\frac{3}{2} \\ 4 \\ -\frac{9}{2} \end{bmatrix}.$$

In the solution to Example 1, notice that $C_{E,B}$ is the inverse of the matrix appearing to the left of the partition in the original augmented matrix; that is, $C_{E,B} = M_B^{-1}$. Taking the inverse of both matrices, we see that $C_{B,E} = M_B$. Of course, this is true in general, because by our boxed procedure, we find $C_{B,E}$ by reducing the matrix $[M_E | M_B] = [I | M_B]$, and this augmented matrix is already in reduced row-echelon form.

In order to find a change-of-coordinates matrix $C_{B,B'}$ for bases B and B' in an n -dimensional vector space V other than $\mathbb{R}^n$, we choose a convenient ordered basis for V and coordinatize the vectors in V relative to that basis, making V look just like $\mathbb{R}^n$. We illustrate this procedure with the vector space P_2 of polynomials of degree at most 2, showing how the work can be carried out with coordinate vectors in $\mathbb{R}^3$.

EXAMPLE 2 Let $B = (x^2, x, 1)$ and $B' = (x^2 - x, 2x^2 - 2x + 1, x^2 - 2x)$ be ordered bases of the vector space P_2 . Find the change-of-coordinates matrix from B to B' , and use it to find the coordinate vector of $2x^2 + 3x - 1$ relative to B' .

SOLUTION Let us use the ordered basis $B = (x^2, x, 1)$ to coordinatize polynomials in P_2 . Identifying each polynomial $a_2x^2 + a_1x + a_0$ with its coordinate vector $[a_2, a_1, a_0]$ relative to the basis B , we obtain the following correspondence from P_2 to $\mathbb{R}^3$:

	Polynomial Basis in P_2	Coordinate Basis in $\mathbb{R}^3$
Old	$B = (x^2, x, 1)$	$([1, 0, 0], [0, 1, 0], [0, 0, 1])$
New	$B' = (x^2 - x, 2x^2 - 2x + 1, x^2 - 2x)$	$([1, -1, 0], [2, -2, 1], [1, -2, 0])$

Working with coordinate vectors in $\mathbb{R}^3$, we compute the desired change-of-coordinates matrix, as described in the box preceding Example 1.

$$\begin{array}{c} \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ -1 & -2 & -2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \\ \text{New basis} \quad \text{Old basis} \\ \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 1 & -2 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 & 0 \end{array} \right] \end{array}$$

The change-of-coordinates matrix is

$$C_{B,B'} = \begin{bmatrix} 2 & 1 & -2 \\ 0 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}.$$

We compute the coordinate vector of $\mathbf{v} = 2x^2 + 3x - 1$ relative to B' , using $C_{B,B'}$ and the coordinate vector of $\mathbf{v}$ relative to B , as follows:

$$\begin{array}{c} \left[\begin{array}{ccc|c} 2 & 1 & -2 \\ 0 & 0 & 1 \\ -1 & -1 & 0 \end{array} \right] \left[\begin{array}{c} 2 \\ 3 \\ -1 \end{array} \right] = \left[\begin{array}{c} 9 \\ -1 \\ -5 \end{array} \right] \\ C_{B,B'} \quad \mathbf{v}_B \quad \mathbf{v}_{B'} \end{array}$$

As a check, we have

$$\underset{\mathbf{v}}{2x^2 + 3x - 1} = 9(\underset{\mathbf{b}_1'}{x^2 - x}) - 1(\underset{\mathbf{b}_2'}{2x^2 - 2x + 1}) - 5(\underset{\mathbf{b}_3'}{x^2 - 2x}).$$

There is another way besides $M_B^{-1}M_{B'}$ to express the change-of-coordinates matrix $C_{B,B'}$. Recall that the coordinate vector $(\mathbf{b}_j)_{B'}$ of $\mathbf{b}_j$ relative to B' is found by reducing the augmented matrix $[M_{B'} | \mathbf{b}_j]$. Thus all n coordinate vectors $(\mathbf{b}_j)_{B'}$ can be found at once by reducing the augmented matrix $[M_{B'} | M_B]$. But this is precisely what our boxed procedure for finding $C_{B,B'}$ calls for. We conclude that

$$C_{B,B'} = \left[\begin{array}{c|c|c|c} | & | & | & | \\ (\mathbf{b}_1)_{B'} & (\mathbf{b}_2)_{B'} & \cdots & (\mathbf{b}_n)_{B'} \\ | & | & | & | \end{array} \right]. \quad (9)$$

Because every n -dimensional vector space V is isomorphic to $\mathbb{R}^n$, Eq. (9) continues to hold for ordered bases B and B' in V . To see explicitly why this is true, we use B' to coordinatize V . Relative to B' the coordinate vectors of vectors in B' are the vectors in the standard ordered basis E of $\mathbb{R}^n$, whereas the coordinate vectors of basis vectors in B are the vectors in the basis $B^* = ((\mathbf{b}_1)_{B'}, (\mathbf{b}_2)_{B'}, \dots, (\mathbf{b}_n)_{B'})$ of $\mathbb{R}^n$. Now the change-of-coordinates matrix from B^* to E is precisely the basis matrix $M_{B'}$ on the right-hand side of Eq. (9). Because

V and $\mathbb{R}^n$ are isomorphic, we see (as in Example 2) that this matrix is also the change-of-coordinates matrix from B to B' .

There are times when it is feasible to use Eq. (9) to find the change-of-coordinates matrix—just noticing how the vector $\mathbf{b}_j$ in B can be expressed as a linear combination of vectors in B' to determine the j th column vector of $C_{B,B'}$ rather than actually coordinatizing V and working within $\mathbb{R}^n$. The next example illustrates this.

EXAMPLE 3 Use Eq. (9) to find the change-of-coordinates matrix $C_{B,B'}$ from the basis $B = (x^2 - 1, x^2 + 1, x^2 + 2x + 1)$ to the basis $B' = (x^2, x, 1)$ in the vector space P_2 of polynomials of degree at most 2.

SOLUTION Relative to $B' = (x^2, x, 1)$, we see at once that the coordinate vectors of $x^2 - 1$, $x^2 + 1$, and $x^2 + 2x + 1$ are $[1, 0, -1]$, $[1, 0, 1]$, and $[1, 2, 1]$, respectively. Changing these to column vectors, we obtain

$$C_{B,B'} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}.$$

SUMMARY

Let V be a vector space with ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$.

1. Each vector $\mathbf{v}$ in V has a unique expression as a linear combination

$$\mathbf{v} = r_1 \mathbf{b}_1 + r_2 \mathbf{b}_2 + \cdots + r_n \mathbf{b}_n.$$

The vector $\mathbf{v}_B = [r_1, r_2, \dots, r_n]$ is the *coordinate vector* of $\mathbf{v}$ relative to B . Associating with each vector $\mathbf{v}$ its coordinate vector $\mathbf{v}_B$ *coordinatizes* V , so that V is *isomorphic* to $\mathbb{R}^n$.

2. Let B and B' be ordered bases of V . There exists a unique $n \times n$ matrix $C_{B,B'}$ such that $C_{B,B'} \mathbf{v}_B = \mathbf{v}_{B'}$ for all vectors $\mathbf{v}$ in V . This is the *change-of-coordinates matrix* from B to B' . It can be computed by coordinatizing V and then applying the boxed procedure on page 391. Alternatively, it can be computed using Eq. (9).

EXERCISES

Exercises 1–7 are a review of Section 3.3. In Exercises 1–6, find the coordinate vector of the given vector relative to the given ordered basis.

1. $\cos 2x$ in $\text{sp}(\sin^2 x, \cos^2 x)$ relative to $(\sin^2 x, \cos^2 x)$
2. $x^3 + x^2 - 2x + 4$ in P_3 relative to $(1, x^2, x, x^3)$.

3. $x^3 + 3x^2 - 4x + 2$ in P_3 relative to $(x, x^2 - 1, x^3, 2x^2)$
4. $x + x^4$ in P_4 relative to $(1, 2x - 1, x^3 + x^4, 2x^3, x^2 + 2)$

5. $\begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ in M_2 relative to

$$\left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \right)$$

6. $\sinh x$ in $\text{sp}(e^x, e^{-x})$ relative to (e^x, e^{-x})
 7. Find the polynomial in P_2 whose coordinate vector relative to the ordered basis $B = (x + x^2, x - x^2, 1 + x)$ is $[3, 1, 2]$.
 8. Let B be an ordered basis for $\mathbb{R}^3$. If

$$C_{E,B} = \begin{bmatrix} 3 & 1 & 2 \\ 4 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix},$$

find the coordinate vector $\mathbf{v}_B$.

9. Let V be a vector space with ordered bases B and B' . If

$$C_{B,B'} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -2 \\ -1 & 0 & 1 \end{bmatrix} \text{ and}$$

$$\mathbf{v} = 3\mathbf{b}_1 - 2\mathbf{b}_2 + \mathbf{b}_3,$$

find the coordinate vector $\mathbf{v}_{B'}$.

In Exercises 10–14, find the change-of-coordinates matrix (a) from B to B' , and (b) from B' to B . Verify that these matrices are inverses of each other.

10. $B = ([1, 1], [1, 0])$ and $B' = ([0, 1], [1, 1])$ in $\mathbb{R}^2$
 11. $B = ([2, 3, 1], [1, 2, 0], [2, 0, 3])$ and $B' = ([1, 0, 0], [0, 1, 0], [0, 0, 1])$ in $\mathbb{R}^3$
 12. $B = ([1, 0, 1], [1, 1, 0], [0, 1, 1])$ and $B' = ([0, 1, 1], [1, 1, 0], [1, 0, 1])$ in $\mathbb{R}^3$
 13. $B = ([1, 1, 1, 1], [1, 1, 1, 0], [1, 1, 0, 0], [1, 0, 0, 0])$ and the standard ordered basis $B' = E$ for $\mathbb{R}^4$
 14. $B = (\sinh x, \cosh x)$ and $B' = (e^x, e^{-x})$ in $\text{sp}(\sinh x, \cosh x)$
 15. Find the change-of-coordinates matrix from B' to B for the bases $B = (x^2, x, 1)$ and $B' = (x^2 - x, 2x^2 - 2x + 1, x^2 - 2x)$ of P_2 in Example 2. Verify that this matrix is the inverse of the change-of-coordinates matrix from B to B' found in that example.

16. Proceeding as in Example 2, find the change-of-coordinates matrix from $B = (x^3, x^2, x, 1)$ to $B' = (x^3 - x^2, x^2 - x, x - 1, x^3 + 1)$ in the vector space P_3 of polynomials of degree at most 3.
 17. Find the change-of-coordinates matrix from B' to B for the vector space and ordered bases given in Exercise 16. Show that this matrix is the inverse of the matrix found in Exercise 16.

In Exercises 18–21, use Eq. (9) in the text to find the change-of-coordinates matrix from B to B' for the given vector space V with ordered bases B and B' .

18. $V = P_3$, B is the basis in Exercise 3, $B' = (x_3, x_2, x, 1)$
 19. $V = P_3$, $B = (x^3 + x^2 + 1, 2x^3 - x^2 + x + 1, x^3 - x + 1, x^2 + x + 1)$, $B' = (x^3, x^2, x, 1)$
 20. V is the space M_2 of all 2×2 matrices, B is the basis in Exercise 5,

$$B' = \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right)$$

21. V is the subspace $\text{sp}(\sin^2 x, \cos^2 x)$ of the vector space F of all real-valued functions with domain $\mathbb{R}$, $B = (\cos 2x, 1)$, $B' = (\sin^2 x, \cos^2 x)$
 22. Find the change-of-coordinates matrix from B' to B for the vector space V and ordered bases B and B' of Exercise 21.
 23. Let B and B' be ordered bases for $\mathbb{R}^n$. Mark each of the following True or False.
 — a. Every change-of-coordinates matrix is square.
 — b. Every $n \times n$ matrix is a change-of-coordinates matrix relative to some bases of $\mathbb{R}^n$.
 — c. If B and B' are orthonormal bases, then $C_{B,B'}$ is an orthogonal matrix.
 — d. If $C_{B,B'}$ is an orthogonal matrix, then B and B' are orthonormal bases.
 — e. If $C_{B,B'}$ is an orthogonal matrix and B is an orthonormal basis, then B' is an orthonormal basis.
 — f. For all choices of B and B' , we have $\det(C_{B,B'}) = 1$.

- g. For all choices of B and B' , we have $\det(C_{B,B'}) \neq 0$.
- h. $\det(C_{B,B'}) = 1$ if and only if $B = B'$.
- i. $C_{B,B'} = I$ if and only if $B = B'$.
- j. Every invertible $n \times n$ matrix is the change-of-coordinates matrix $C_{B,B'}$ for some ordered bases B and B' of $\mathbb{R}^n$.
24. For ordered bases B and B' in $\mathbb{R}^n$, explain how the change-of-coordinates matrix from B to B' is related to the change-of-coordinates matrices from B to E and from E to B' .
25. Let B, B' , and B'' be ordered bases for $\mathbb{R}^n$. Find the change-of-coordinates matrix from B to B'' in terms of $C_{B,B'}$ and $C_{B',B''}$. [Hint: For a vector v in $\mathbb{R}^n$, what matrix times v_B gives $v_{B''}$?

7.2

MATRIX REPRESENTATIONS AND SIMILARITY

Let V and V' be vector spaces with ordered bases $B = (b_1, b_2, \dots, b_n)$ and $B' = (b'_1, b'_2, \dots, b'_m)$, respectively. If $T: V \rightarrow V'$ is a linear transformation, then Theorem 3.10 shows that the *matrix representation of T relative to B, B'* , which we now denote as $R_{B,B'}$, is given by

$$R_{B,B'} = \begin{bmatrix} | & | & & | \\ T(b_1)_{B'} & T(b_2)_{B'} & \cdots & T(b_n)_{B'} \\ | & | & & | \end{bmatrix}, \quad (1)$$

where $T(b_j)_{B'}$ is the coordinate vector of $T(b_j)$ relative to B' . Furthermore, $R_{B,B'}$ is the *unique* matrix satisfying

$$T(v)_{B'} = R_{B,B'} v_B \quad \text{for all } v \text{ in } V. \quad (2)$$

Let us denote by $M_{T(B)}$ the $m \times n$ matrix whose j th column vector is $T(b_j)$. From Eq. (1), we see that we can compute $R_{B,B'}$ by row-reducing the augmented matrix $[M_{T(B)} | I_m]$ to obtain $[I_m | R_{B,B'}]$. The discussion surrounding Theorem 3.10 shows how computations involving $T: V \rightarrow V'$ can be carried out in $\mathbb{R}^n$ and $\mathbb{R}^m$ using $R_{B,B'}$ and coordinates of vectors relative to the basis B of V and B' of V' , in view of the isomorphisms of V with $\mathbb{R}^n$ and of V' with $\mathbb{R}^m$ that these coordinatizations provide.

It is the purpose of this section to study the effect that choosing different bases for coordinatization has on the matrix representations of a linear transformation. For simplicity, we shall derive our results in terms of the vector spaces $\mathbb{R}^n$. They can then be carried over to other finite-dimensional vector spaces using coordinatization isomorphisms.

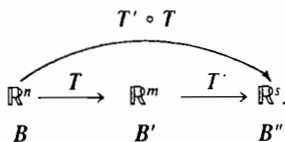
The Multiplicative Property of Matrix Representations

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T': \mathbb{R}^m \rightarrow \mathbb{R}^l$ be linear transformations. Section 2.3 showed that composition of linear transformations corresponds to multiplication

cf. their standard matrix representations; that is, for standard bases, we have

$$\text{Matrix for } (T' \circ T) = (\text{Matrix for } T')(\text{Matrix for } T).$$

The analogous property holds for representations relative to any bases. Consider the linear transformations and ordered bases shown by the diagram



The following vector and matrix counterpart diagram shows the action of the transformations on vectors:

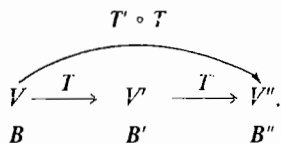
$$\begin{array}{ccccc} & & R_{R, R'} & & \\ & \curvearrowright & & \curvearrowleft & \\ \mathbf{v}_B & \xrightarrow{R_{R, B'}} & T(\mathbf{v})_{B'} & \xrightarrow{R_{B', B''}} & T'(T(\mathbf{v}))_{B''} \\ & R_{R, B'} & & R_{B', B''} & \end{array} \quad (3)$$

The matrix $R_{B, B'}$ under the first arrow transforms the vector on the left of the arrow into the vector on the right by left multiplication, as indicated by Eq. (2). The matrix under the second arrow acts in a similar way. Thus the matrix product $R_{B', B''} R_{B, B'}$ transforms $\mathbf{v}_B$ into $T'(T(\mathbf{v}))_{B''}$. However, the matrix representation $R_{B, B''}$ of $T' \circ T$ relative to B, B'' is the *unique* matrix that carries $\mathbf{v}_B$ into $T'(T(\mathbf{v}))_{B''}$ for all $\mathbf{v}_B$ in $\mathbb{R}^n$. Thus we must have

$$R_{B, B''} = R_{B', B''} R_{B, B'}. \quad (4)$$

Notice that the order of the matrices in this product is *opposite* to the order in which they appear in diagram (3). (See the footnote on page 160 to see why this is so.)

Equation (4) holds equally well for matrix representations of linear transformations of general finite-dimensional vector spaces, as shown in the diagram



EXAMPLE 1 Let $B = (x^4, x^3, x^2, x, 1)$, which is an ordered basis for P_4 , and let $T: P_4 \rightarrow P_4$ be the differentiation transformation. Find the matrix representation $R_{B, B}$ of T relative to B, B , and illustrate how it can be used to differentiate the polynomial function

$$3x^4 - 5x^3 + 7x^2 - 8x + 2.$$

Then use calculus to show that $(R_{B, B})^5 = O$.

SOLUTION Because $T(x^4) = kx^{4-1}$, we see that the matrix representation $R_{B,B}$ in Eq. (1) is

$$R_{B,B} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Now the coordinate vector of $p(x) = 3x^4 - 5x^3 + 7x^2 - 8x + 2$ relative to B is

$$p(x)_B = \begin{bmatrix} 3 \\ -5 \\ 7 \\ -8 \\ 2 \end{bmatrix}.$$

Applying Eq. (2) with $B' = B$, we find that

$$T(p(x))_B = R_{B,B}p(x)_B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -5 \\ 7 \\ -8 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 12 \\ -15 \\ 14 \\ -8 \end{bmatrix}.$$

Thus, $p'(x) = T(p(x)) = 12x^3 - 15x^2 + 14x - 8$.

The discussion surrounding Eq. (4) shows that the linear transformation of P_4 that computes the fifth derivative has matrix representation $(R_{B,B})^5$. Because the fifth derivative of any polynomial in P_4 is zero, we see that $(R_{B,B})^5 = O$. ■

Similarity of Representations Relative to Different Bases

Let B and B' be ordered bases for $\mathbb{R}^n$. In the preceding section we saw that the change-of-coordinates matrix $C_{B,B'}$ from B to B' is the unique matrix satisfying

$$\mathbf{v}_{B'} = C_{B,B'} \mathbf{v}_B \quad \text{for all } \mathbf{v} \text{ in } \mathbb{R}^n.$$

Comparison of this equation with Eq. (2) shows that $C_{B,B'}$ is the matrix representation, relative to B, B' , of the identity transformation $\iota: \mathbb{R}^n \rightarrow \mathbb{R}^n$, where $\iota(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$.

The relationship between the matrix representations R_B and $R_{B'}$ for the linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ can be derived from the following diagram:

$$\begin{array}{ccccccc} & & & T & & & \\ & \swarrow & & \searrow & & \swarrow & \\ \mathbb{R}^n & \xrightarrow{\iota} & \mathbb{R}^n & \xrightarrow{T} & \mathbb{R}^n & \xrightarrow{\iota} & \mathbb{R}^n \\ B' & & B & & B & & B' \end{array}$$

The vector and matrix counterpart diagram similar to diagram (3) is

$$\begin{array}{ccccccc}
 & & & R_{B'} & & & \\
 & & \text{---} & \text{---} & \text{---} & \text{---} & \\
 v_{B'} & \xrightarrow{C_{B',B}} & v_B & \xrightarrow{R_B} & T(v)_B & \xrightarrow{C_{B,B'}} & T(v)_{B'}
 \end{array}$$

Remembering to reverse the order, we find that

$$R_{B'} = C_{B,B'} R_B C_{B',B}. \quad (5)$$

Reading the matrix product in Eq. (5) from right to left, we see that in order to compute $T(v)_{B'}$ from $v_{B'}$ if we know R_B , we:

1. change from B' to B coordinates,
2. compute the transformation relative to B coordinates,
3. change back to B' coordinates.

Equation (5) makes this procedure easy to remember.

The matrices $C_{B,B'}$ and $C_{B',B}$ are inverses of each other. If we drop subscripts for a moment and let $C = C_{B',B}$, then Eq. (5) becomes

$$R_{B'} = C^{-1} R_B C. \quad (6)$$

Recall that two $n \times n$ matrices A and R are *similar* if there exists an invertible $n \times n$ matrix C such that $R = C^{-1}AC$. (See Definition 5.4.) We have shown that matrix representations of the same transformation relative to different bases are similar. We state this as a theorem in the context of a general finite-dimensional vector space.

THEOREM 7.1 Similarity of Matrix Representations of T

Let T be a linear transformation of a finite-dimensional vector space V into itself, and let B and B' be ordered bases of V . Let R_B and $R_{B'}$ be the matrix representations of T relative to B and B' , respectively. Then

$$R_{B'} = C^{-1} R_B C,$$

where $C = C_{B',B}$ is the change-of-coordinates matrix from B' to B . Consequently, $R_{B'}$ and R_B are similar matrices.

EXAMPLE 2 Consider the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T(x_1, x_2, x_3) = (x_1 + x_2 + x_3, x_1 + x_2, x_3).$$

Find the standard matrix representation A of T , and find the matrix representation R_B of T relative to B , where

$$B = ([1, 1, 0], [1, 0, 1], [0, 1, 1]).$$

In addition, find an invertible matrix C such that $R_B = C^{-1}AC$.

SOLUTION Here the standard ordered basis E plays the role that the basis B played in Theorem 7.1, and the basis B here plays the role played by B' in that theorem. Of course, the standard matrix representation of T is

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We compute the matrix representation R_B by reducing $[M_B | M_{T(B)}]$ to $[I | R_B]$ as follows:

$$\begin{array}{c} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 0 & 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 2 & 2 & 2 \\ 0 & -1 & 1 & 0 & -1 & -1 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{array} \right] \\ \begin{array}{cccccc} b_1 & b_2 & b_3 & T(b_1) & T(b_2) & T(b_3) \end{array} \\ \\ \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 & 1 \\ 0 & 0 & 2 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{array} \right] \\ \begin{array}{cccccc} & & & I & & R_B \end{array} \end{array}$$

Thus,

$$R_B = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

An invertible matrix C such that $R_B = C^{-1}AC$ is $C = C_{B,E}$. We find $C = C_{B,E}$ by reducing the augmented matrix $[M_E | M_B] = [I | M_B]$. Because this matrix is already reduced, we see that

$$C = M_B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

in this case. The matrices A and R_B are similar matrices, both representing the given linear transformation. As a check, we could compute that $AC = CR_B$. ■

We give an example illustrating Theorem 7.1 in the case of a finite-dimensional vector space other than $\mathbb{R}^n$.

EXAMPLE 3 For the space P_2 of polynomials of degree at most 2, let $T: P_2 \rightarrow P_2$ be defined by $T(p(x)) = p(x-1)$. Consider the two ordered bases $B = (x^2, x, 1)$ and $B' = (x, x+1, x^2-1)$. Find the matrix representations R_B and $R_{B'}$ of T and a matrix C such that $R_{B'} = C^{-1}R_B C$.

SOLUTION Because

$$\begin{aligned} T(x^2) &= (x-1)^2 = x^2 - 2x + 1, \\ T(x) &= x - 1, \\ T(1) &= 1, \end{aligned}$$

the matrix representation of T relative to $B = \{x^2, x, 1\}$ is

$$R_B = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}.$$

Next we compute the change-of-coordinates matrices $C_{B,B'}$ and $C_{B',B}$. We see that

$$C = C_{B',B} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}.$$

Moreover

$$\begin{aligned} x^2 &= -1(x) + 1(x+1) + 1(x^2-1), \\ x &= 1(x) + 0(x+1) + 0(x^2-1), \\ 1 &= -1(x) + 1(x+1) + 0(x^2-1), \end{aligned}$$

so

$$C_{B,B'} = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Notice that $C_{B,B'}$ can be computed as $(C_{B',B})^{-1}$. We now have

$$\begin{aligned} R_{B'} &= \underbrace{\begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}_{C_{B,B'}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}}_{R_B} \underbrace{\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}}_{C_{B',B}} \\ &= \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & -2 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & -3 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

Alternatively, $R_{B'}$ can be computed directly as

$$R_{B'} = \begin{bmatrix} | & | & | \\ T(\mathbf{b}'_1)_{B'} & T(\mathbf{b}'_2)_{B'} & T(\mathbf{b}'_3)_{B'} \\ | & | & | \end{bmatrix}.$$

We have seen that matrix representations of the same transformation relative to different bases are similar. Conversely, any two similar matrices can be viewed as representations of the same transformation relative to different bases. To see this, let A be an $n \times n$ matrix, and let C be any invertible $n \times n$ matrix. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $T(\mathbf{x}) = A\mathbf{x}$ so that A is the standard matrix representation of T . Because C is invertible, its column vectors are independent and form a basis for $\mathbb{R}^n$. Let B be the ordered basis having as j th vector the

j th column vector of C . Then C is precisely the change-of-coordinates matrix from B to the standard ordered basis E . That is, $C = C_{B,E}$. Consequently, $C^{-1}AC = C_{E,B}AC_{B,E}$ is the matrix representation of T relative to B .

Significance of the Similarity Relationship for Matrices

Two $n \times n$ matrices are similar if and only if they are matrix representations of the same linear transformation T relative to suitable ordered bases.

The Interplay of Matrices and Linear Transformations

Let V be an n -dimensional vector space, and let $T: V \rightarrow V$ be a linear transformation. Suppose that T has a property that can be characterized in terms of the mapping, without reference to coordinates relative to any basis. Assertions about T of the form

- There is a basis of eigenvectors of T ,
- The nullspace of T has dimension 4,
- The eigenspace E_λ of T has dimension 2,

are *coordinate-independent* assertions. For another example, T has λ as an eigenvalue if and only if

$$T(\mathbf{v}) = \lambda \mathbf{v} \quad \text{for some nonzero vector } \mathbf{v} \text{ in } V. \quad (7)$$

This statement makes no reference to coordinates relative to a basis. The existence and value of an eigenvalue λ is coordinate independent. Of course, the existence and the characterization of eigenvectors $\mathbf{v}$ corresponding to λ as given in Eq. (7) are also coordinate independent. However, an eigenvector $\mathbf{v}$ can have different coordinates relative to different bases B and B' of V , so the matrix representations R_B and $R_{B'}$ of T may have different eigenvectors. Equation (7) expressed in terms of the coordinatizations of V by B and B' becomes

$$R_B(\mathbf{v}_B) = \lambda \mathbf{v}_B \quad \text{and} \quad R_{B'}(\mathbf{v}_{B'}) = \lambda \mathbf{v}_{B'},$$

respectively. While the coordinates of the eigenvector $\mathbf{v}$ change, the value λ doesn't change. That is, if λ is an eigenvalue of T , then λ is an eigenvalue of every matrix representation of T .

Now any two similar matrices can be viewed as matrix representations of the same linear transformation T , relative to suitable bases. Consequently, similar matrices must share any properties that can be described for the transformation in a coordinate-free fashion. In particular we obtain, with no additional work, this nice result:

Similar matrices have the same eigenvalues.

We take a moment to expand on the ideas we have just introduced. Let V be a vector space of finite dimension n . The study of linear transformations $T: V \rightarrow V$ is essentially the same as the study of products Ax of vectors x in $\mathbb{R}^n$ by $n \times n$ matrices A . We should understand this relationship thoroughly and be able to bounce back and forth at will from $T(v)$ for v in V to Ax for x in $\mathbb{R}^n$. Sometimes a theorem that is not immediately obvious from one point of view is quite easy from another. For example, from the matrix point of view, it is not immediately apparent that similar matrices have the same eigenvalues; we ask for a matrix proof in Exercise 27. However, the truth of this statement becomes obvious from the linear transformation point of view. On the other hand, it is not obvious from Eq. (7) concerning an eigenvalue of a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ that a linear transformation of V can have at most n eigenvalues. However, this is easy to establish from the matrix point of view: $Ax = \lambda x$ has a nontrivial solution if and only if the coefficient matrix $A - \lambda I$ of the system $(A - \lambda I)x = 0$ has determinant zero, and $\det(A - \lambda I) = 0$ is a polynomial equation of degree n .

We now state a theorem relating eigenvalues and eigenvectors of similar matrices. Exercises 24 and 25 ask for proofs of the second and third statements in the theorem. We have already proved the first statement.

THEOREM 7.2 Eigenvalues and Eigenvectors of Similar Matrices

Let A and R be similar $n \times n$ matrices, so that $R = C^{-1}AC$ for some invertible $n \times n$ matrix C . Let the eigenvalues of A be the (not necessarily distinct) numbers $\lambda_1, \lambda_2, \dots, \lambda_n$.

1. The eigenvalues of R are also $\lambda_1, \lambda_2, \dots, \lambda_n$.
2. The algebraic and geometric multiplicity of each λ_i as an eigenvalue of A remains the same as when it is viewed as an eigenvalue of R .
3. If v_i in $\mathbb{R}^n$ is an eigenvector of the matrix A corresponding to λ_i , then $C^{-1}v_i$ is an eigenvector of the matrix R corresponding to λ_i .

We give one more illustration of the usefulness of this interplay between linear transformations and matrices, and of the significance of similarity. Let V be a finite-dimensional vector space, and let $T: V \rightarrow V$ be a linear transformation. It was easy to define the notions of *eigenvalue* and *eigenvector* in terms of T . We can now define the **geometric multiplicity** of an eigenvalue λ to be the dimension of the eigenspace $E_\lambda = \{v \in V \mid T(v) = \lambda v\}$. However, at this time there is no obvious way to characterize the algebraic multiplicity of λ exclusively in terms of the mapping T , without coordinatization. Consequently, we define the **algebraic multiplicity** of λ to be its algebraic multiplicity

as an eigenvalue of a matrix representation of λ . This makes sense because this algebraic multiplicity of λ is the same for *all* matrix representations of T . Statement 2 of Theorem 7.2 assures us that this is the case.

Diagonalization

Let $T: V \rightarrow V$ be a linear transformation of an n -dimensional vector space into itself. Suppose that there exists an ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ of V composed of eigenvectors of T . Let the eigenvalue corresponding to $\mathbf{b}_i$ be λ_i . Then the matrix representation of T relative to B has the simple diagonal form

$$D = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}.$$

We give a definition of diagonalization for linear transformations that clearly parallels one for matrices.

DEFINITION 7.2 Diagonalizable Transformation

A linear transformation T of a finite-dimensional vector space V into itself is **diagonalizable** if V has an ordered basis consisting of eigenvectors of T .

EXAMPLE 4 Show that reflection of the plane $\mathbb{R}^2$ in the line $y = mx$ is a diagonalizable transformation and find a diagonal matrix representation for it.

SOLUTION Reflection of the plane $\mathbb{R}^2$ in a line through the origin is a linear transformation. (See Illustration 2 on page 297.) Figure 7.1 shows that $\mathbf{b}_1 = [1, m]$ is carried into itself and $\mathbf{b}_2 = [-m, 1]$ is carried into $-\mathbf{b}_2$. Thus $\mathbf{b}_1$ is an eigenvector of this transformation with corresponding eigenvalue 1 whereas $\mathbf{b}_2$

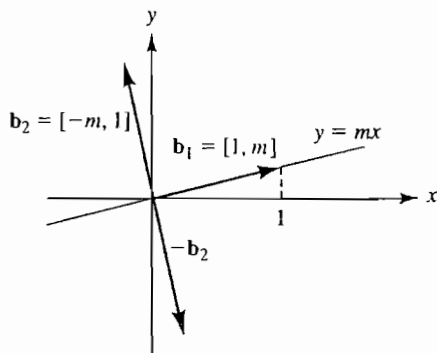


FIGURE 7.1
Reflection in the line $y = mx$.

is an eigenvector with eigenvalue -1 . Because $\{\mathbf{b}_1, \mathbf{b}_2\}$ is a basis of eigenvectors for $\mathbb{R}^2$, the reflection transformation is diagonalizable. Relative to the ordered basis $B = (\mathbf{b}_1, \mathbf{b}_2)$,

$$R_B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

If V has a known ordered basis B of eigenvectors, it becomes easy to compute the k -fold composition $T^k(\mathbf{v})$ for any positive integer k and for any vector $\mathbf{v}$ in V . We need only find the coordinate vector $\mathbf{d}$ in $\mathbb{R}^n$ of $\mathbf{v}$ relative to B , so that

$$\mathbf{v} = d_1\mathbf{b}_1 + d_2\mathbf{b}_2 + \cdots + d_n\mathbf{b}_n.$$

Then

$$T^k(\mathbf{v}) = d_1\lambda_1^k\mathbf{b}_1 + d_2\lambda_2^k\mathbf{b}_2 + \cdots + d_n\lambda_n^k\mathbf{b}_n. \quad (8)$$

Of course, this is the transformation analogue of the computation of $A^k\mathbf{x}$ discussed in Section 5.1. We illustrate with an example.

EXAMPLE 5 Consider the vector space P_2 of polynomials of degree at most 2, and let B' be the ordered basis $(1, x, x^2)$ for P_2 . Let $T: P_2 \rightarrow P_2$ be the linear transformation such that

$$T(1) = 3 + 2x + x^2, \quad T(x) = 2, \quad T(x^2) = 2x^2.$$

Find $T^4(x + 2)$.

SOLUTION The matrix representation of T relative to B' is

$$R_{B'} = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 0 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

Using the methods of Chapter 5, we easily find the eigenvalues and eigenvectors of $R_{B'}$ and of T given in Table 7.1.

TABLE 7.1

Eigenvalues	Eigenvectors of $R_{B'}$	Eigenvectors of T
$\lambda_1 = -1$	$\mathbf{w}_1 = [-3, 6, 1]$	$p_1(x) = -3 + 6x + x^2$
$\lambda_2 = 2$	$\mathbf{w}_2 = [0, 0, 1]$	$p_2(x) = x^2$
$\lambda_3 = 4$	$\mathbf{w}_3 = [2, 1, 1]$	$p_3(x) = 2 + x + x^2$

Let B be the ordered basis $(-3 + 6x + x^2, x^2, 2 + x + x^2)$ consisting of these eigenvectors. We can find the coordinate vector $\mathbf{d}$ of $x + 2$ relative to the basis B by inspection. Because

$$x + 2 = 0(x^2 + 6x - 3) + (-1)x^2 + 1(x^2 + x + 2).$$

we see that

$$d_1 = 0, \quad d_2 = -1, \quad \text{and} \quad d_3 = 1.$$

Thus, Eq. (8) has the form

$$T^k(x+2) = 2^k(-1)x^2 + 4^k(1)(x^2 + x + 2).$$

In particular,

$$T^4(x+2) = -16x^2 + 256(x^2 + x + 2) = 240x^2 + 256x + 512. \quad \blacksquare$$

SUMMARY

Let $T: V \rightarrow V$ be a linear transformation of a finite-dimensional vector space into itself.

1. If B and B' are ordered bases of V , then the matrix representations R_B and $R_{B'}$ of T relative to B and to B' are similar. That is, there is an invertible matrix C —namely, $C = C_{B', B}$ —such that

$$R_{B'} = C^{-1}R_B C.$$

2. Conversely, two similar $n \times n$ matrices represent the same linear transformation of $\mathbb{R}^n$ into $\mathbb{R}^n$ relative to two suitably chosen ordered bases.
3. Similar matrices have the same eigenvalues with the same algebraic and geometric multiplicities.
4. If A and R are similar matrices with $R = C^{-1}AC$, and if $\mathbf{v}$ is an eigenvector of A , then $C^{-1}\mathbf{v}$ is an eigenvector of R corresponding to the same eigenvalue.
5. The transformation T is diagonalizable if V has a basis B consisting of eigenvectors of T . In this case, the matrix representation R_B is a diagonal matrix and computation of $T^k(\mathbf{v})$ by $R_B^k(\mathbf{v}_B)$ becomes relatively easy.

EXERCISES

In Exercises 1–14, find the matrix representations R_B and $R_{B'}$ and an invertible matrix C such that $R_{B'} = C^{-1}R_B C$ for the linear transformation T of the given vector space with the indicated ordered bases B and B' .

1. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T([x, y]) = [x - y, x + 2y]$; $B = ([1, 1], [2, 1])$, $B' = E$
2. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T([x, y]) = [2x + 3y, x + 2y]$; $B = ([1, -1], [1, 1])$, $B' = ([2, 3], [1, 2])$
3. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T([x, y, z]) = [x + y, x + z, y - z]$; $B = ([1, 1, 1], [1, 1, 0], [1, 0, 0])$, $B' = E$
4. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T([x, y, z]) = [5x, 2y, 3z]$; B and B' as in Exercise 3
5. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T([x, y, z]) = [z, 0, x]$; $B = ([3, 1, 2], [1, 2, 1], [2, -1, 0])$, $B' = ([1, 2, 1], [2, 1, -1], [5, 4, 1])$
6. $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as reflection of the plane through the line $5x = 3y$; $B = ([3, 5], [5, -3])$, $B' = E$

7. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as reflection of $\mathbb{R}^3$ through the plane $x + y + z = 0$; $B = ([1, 0, -1], [1, -1, 0], [1, 1, 1])$, $B' = E$
8. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as reflection of $\mathbb{R}^3$ through the plane $2x + 3y + z = 0$; $B = ([2, 3, 1], [0, 1, -3], [1, 0, -2])$, $B' = E$
9. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as projection on the plane $x_2 = 0$; $B = E$, $B' = ([1, 0, 1], [1, 0, -1], [0, 1, 0])$
10. $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as projection on the plane $x + y + z = 0$; B and B' as in Exercise 7.
11. $T: P_2 \rightarrow P_2$ defined by $T(p(x)) = p(x+1) + p(x)$; $B = (x^2, x, 1)$, $B' = (1, x, x^2)$
12. $T: P_2 \rightarrow P_2$ as in Exercise 11, but using $B = (x^2, x, 1)$ and $B' = (x^2 + 1, x + 1, 2)$
13. $T: P_3 \rightarrow P_3$ defined by $T(p(x)) = p'(x)$, the derivative of $p(x)$; $B = (x^3, x^2, x, 1)$, $B' = (1, x + 1, x^2 + 1, x^3 + 1)$
14. $T: W \rightarrow W$, where $W = \text{sp}(e^x, xe^x)$ and T is the derivative transformation; $B = (e^x, xe^x)$, $B' = (2xe^x, 3e^x)$
15. Let $T: P_2 \rightarrow P_2$ be the linear transformation and B' the ordered basis of P_2 given in Example 3. Find the matrix representation $R_{B'}$ of T by computing the matrix with column vectors $T(b'_1)_{B'}$, $T(b'_2)_{B'}$, $T(b'_3)_{B'}$.
16. Repeat Exercise 15 for the transformation in Example 5 and the basis $B' = (x + 1, -1, x^2 + x)$ of P_2 .

In Exercises 17–22, find the eigenvalues λ_i and the corresponding eigenspaces of the linear transformation T . Determine whether the linear transformation is diagonalizable.

17. T defined on $\mathbb{R}^2$ by $T([x, y]) = [2x - 3y, -3x + 2y]$
18. T defined on $\mathbb{R}^2$ by $T([x, y]) = [x - y, -x + y]$
19. T defined on $\mathbb{R}^3$ by $T([x_1, x_2, x_3]) = [x_1 + x_3, x_2, x_1 + x_3]$

20. T defined on $\mathbb{R}^3$ by $T([x_1, x_2, x_3]) = [x_1, 4x_2 + 7x_3, 2x_2 - x_3]$
21. T defined on $\mathbb{R}^3$ by $T([x_1, x_2, x_3]) = [5x_1, -5x_1 + 3x_2 - 5x_3, -3x_1 - 2x_2]$
22. T defined on $\mathbb{R}^3$ by $T([x_1, x_2, x_3]) = [3x_1 - x_2 + x_3, -2x_1 + 2x_2 - x_3, 2x_1 + x_2 + 4x_3]$
23. Mark each of the following True or False
 - a. Two similar $n \times n$ matrices represent the same linear transformation of $\mathbb{R}^n$ into itself relative to the standard basis.
 - b. Two different $n \times n$ matrices represent different linear transformations of $\mathbb{R}^n$ into itself relative to the standard basis.
 - c. Two similar $n \times n$ matrices represent the same linear transformation of $\mathbb{R}^n$ into itself relative to two suitably chosen bases for $\mathbb{R}^n$.
 - d. Similar matrices have the same eigenvalues and eigenvectors.
 - e. Similar matrices have the same eigenvalues with the same algebraic and geometric multiplicities.
 - f. If A and C are $n \times n$ matrices and C is invertible and v is an eigenvector of A , then $C^{-1}v$ is an eigenvector of $C^{-1}AC$.
 - g. If A and C are $n \times n$ matrices and C is invertible and v is an eigenvector of A , then Cv is an eigenvector of CAC^{-1} .
 - h. Any two $n \times n$ diagonal matrices are similar.
 - i. Any two $n \times n$ diagonalizable matrices having the same eigenvectors are similar.
 - j. Any two $n \times n$ diagonalizable matrices having the same eigenvalues of the same algebraic multiplicities are similar.
24. Prove statement 2 of Theorem 7.2.
25. Prove statement 3 of Theorem 7.2.
26. Let A and R be similar matrices. Prove in two ways that A^2 and R^2 are similar matrices: using a matrix argument, and using a linear transformation argument.
27. Give a determinant proof that similar matrices have the same eigenvalues.

EIGENVALUES: FURTHER APPLICATIONS AND COMPUTATIONS

This chapter deals with further applications of eigenvalues and with the computation of eigenvalues. In Section 8.1, we discuss quadratic forms and their diagonalization. The principal axis theorem (Theorem 8.1) asserts that every quadratic form can be diagonalized. This is probably the most important result in the chapter, having applications to the vibration of elastic bodies, to quantum mechanics, and to electric circuits. Presentations of such applications are beyond the scope of this text, and we have chosen to present a more accessible application in Section 8.2: classification of conic sections and quadric surfaces. Although the facts about conic sections may be familiar to you, their easy derivation from the principal axis theorem should be seen and enjoyed.

Section 8.3 applies the principal axis theorem to local extrema of functions and discusses maximization and minimization of quadratic forms on unit spheres. The latter topic is again important in vibration problems; it indicates that eigenvalues of maximum and of minimum magnitudes can be found for symmetric matrices by using techniques from advanced calculus for finding extrema of quadratic forms on unit spheres.

In Section 8.4, we sketch three methods for computing eigenvalues: the power method, Jacobi's method, and the QR method. We attempt to make as intuitively clear as we can how and why each method works, but proofs and discussions of efficiency are omitted.

This chapter contains some applications to geometry and analysis that are usually phrased in terms of points in $\mathbb{R}^n$ rather than in terms of vectors. We will be studying these applications using our work with vectors. To permit a natural and convenient use of the classical terminology of points while working with vectors, we relax for this chapter our convention that the boldface $\mathbf{x}$ always be a vector $[x_1, x_2, \dots, x_n]$, and allow the same notation to represent the point $(x_1, x_2, \dots, x_n)$ as well.

8.1

DIAGONALIZATION OF QUADRATIC FORMS

Quadratic Forms

A quadratic form in one variable x is a polynomial $f(x) = ax^2$, where $a \neq 0$. A quadratic form in two variables x and y is a polynomial $f(x, y) = ax^2 + bxy + cy^2$, where at least one of a , b , or c is nonzero. The term *quadratic* means *degree 2*. The term *form* means *homogeneous*; that is, each summand contains a product of the *same number* of variables—namely, 2 for a quadratic form. Thus, $3x^2 - 4xy$ is a quadratic form in x and y , but $x^2 + y^2 - 4$ and $x - 3y^2$ are not quadratic forms.

Turning to the general case, a **quadratic form** $f(\mathbf{x}) = f(x_1, x_2, \dots, x_n)$ in n variables is a polynomial that can be written, using summation notation, as

$$f(\mathbf{x}) = \sum_{\substack{i \leq j \\ i, j=1}}^n u_{ij} x_i x_j \quad (1)$$

where not all u_{ij} are zero. To illustrate, the general quadratic form in x_1, x_2 , and x_3 is

$$u_{11}x_1^2 + u_{12}x_1x_2 + u_{13}x_1x_3 + u_{22}x_2^2 + u_{23}x_2x_3 + u_{33}x_3^2. \quad (2)$$

HISTORICAL NOTE In 1826, CAUCHY DISCUSSED QUADRATIC FORMS IN THREE VARIABLES—forms of the type $Ax^2 + By^2 + Cz^2 + 2Dxy + 2Exz + 2Fyz$. He showed that the characteristic equation formed from the determinant

$$\begin{vmatrix} A & D & E \\ D & B & F \\ E & F & C \end{vmatrix}$$

remains the same under any change of rectangular axes, what we would call an *orthogonal coordinate change*. Furthermore, he demonstrated that one could always find axes such that the new form has only the square terms. Three years later, Cauchy generalized the result to quadratic forms in n variables. (The matrices of such forms are $n \times n$ symmetric matrices.) He showed that the roots $\lambda_1, \lambda_2, \dots, \lambda_n$ of the characteristic equation are all real, and he showed how to find the linear substitution that converts the original form to the form $\lambda_1 x_1^2 + \lambda_2 x_2^2 + \dots + \lambda_n x_n^2$. In modern terminology, Cauchy had proved Theorem 5.5, that every real symmetric matrix is diagonalizable. In the two-variable case, Cauchy's proof amounts to finding the maximum and minimum of the quadratic form $f(x, y) = Ax^2 + 2Bxy + Cy^2$ subject to the condition that $x^2 + y^2 = 1$. In geometric terms, the point at which the extreme value occurs is that point on the unit circle which also lies on the end of one axis of one of the family of ellipses (or hyperbolas) described by the quadratic form. If one then takes the line from the origin to that point as one of the axes and the perpendicular to that line as the other, the equation in relation to those axes will have only the squares of the variables, as desired. To determine an extreme point subject to a condition, Cauchy uses what is today called the principle of Lagrange multipliers.

A computation shows that

$$\begin{aligned}
 [x_1, x_2, x_3] \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\
 &= [x_1, x_2, x_3] \begin{bmatrix} u_{11}x_1 + u_{12}x_2 + u_{13}x_3 \\ u_{22}x_2 + u_{23}x_3 \\ u_{33}x_3 \end{bmatrix} \\
 &= x_1(u_{11}x_1 + u_{12}x_2 + u_{13}x_3) + x_2(u_{22}x_2 + u_{23}x_3) + x_3(u_{33}x_3) \\
 &= u_{11}x_1^2 + u_{12}x_1x_2 + u_{13}x_1x_3 + u_{22}x_2^2 + u_{23}x_2x_3 + u_{33}x_3^2,
 \end{aligned}$$

which is again form (2). We can verify that the term involving u_{ij} for $i \leq j$ in the expansion of

$$[x_1, x_2, \dots, x_n] \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad (3)$$

is precisely $u_{ij}x_ix_j$. Thus, matrix product (3) is a 1×1 matrix whose sole entry is equal to sum (1).

Every quadratic form in n variables x_i can be written as $\mathbf{x}^T U \mathbf{x}$, where $\mathbf{x}$ is the column vector of variables and U is a nonzero upper-triangular matrix.

We will call the matrix U the **upper-triangular coefficient matrix** of the quadratic form.

EXAMPLE 1 Write $x^2 - 2xy + 6xz + z^2$ in the form of matrix product (3).

SOLUTION We obtain the matrix expression

$$[x, y, z] \begin{bmatrix} 1 & -2 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Just think of x as the first variable, y as the second, and z as the third. The summand $-2xy$, for example, gives the coefficient -2 in the row 1, column 2 position. ■

A matrix expression for a given quadratic form is by no means unique: $\mathbf{x}^T A \mathbf{x}$ gives a quadratic form for any nonzero $n \times n$ matrix A , and the form can be rewritten as $\mathbf{x}^T U \mathbf{x}$, where U is upper triangular.

EXAMPLE 2 Expand

$$[x, y, z] \begin{bmatrix} -1 & 3 & 1 \\ 2 & 1 & 0 \\ -2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

and find the upper-triangular coefficient matrix for the quadratic form.

SOLUTION We find that

$$\begin{aligned} [x, y, z] \begin{bmatrix} -1 & 3 & 1 \\ 2 & 1 & 0 \\ -2 & 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= [x, y, z] \begin{bmatrix} -x + 3y + z \\ 2x + y \\ -2x + 2y + 4z \end{bmatrix} \\ &= x(-x + 3y + z) + y(2x + y) + z(-2x + 2y + 4z) \\ &= -x^2 + 3xy + xz + 2xy + y^2 - 2xz + 2yz + 4z^2 \\ &= -x^2 + 5xy + y^2 - xz + 2yz + 4z^2. \end{aligned}$$

The upper-triangular coefficient matrix is

$$U = \begin{bmatrix} -1 & 5 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}. \quad \blacksquare$$

All the nice things that we will prove about quadratic forms come from the fact that any quadratic form $f(\mathbf{x})$ can be expressed as $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$, where A is a *symmetric* matrix.

EXAMPLE 3 Find the symmetric coefficient matrix of the form $x^2 - 2xy + 6xz + z^2$ discussed in Example 1.

SOLUTION Rewriting the form as

$$x^2 - xy - yx + 3xz + 3zx + z^2,$$

we obtain the symmetric coefficient matrix

$$A = \begin{bmatrix} 1 & -1 & 3 \\ -1 & 0 & 0 \\ 3 & 0 & 1 \end{bmatrix}. \quad \blacksquare$$

As illustrated in Example 3, we obtain the symmetric matrix A of the form

$$f(\mathbf{x}) = \sum_{\substack{i \leq j \\ i, j=1}}^n u_{ij} x_i x_j \quad (4)$$

by writing each *cross term* $u_{ij} x_i x_j$ as $(u_{ij}/2)x_i x_j + (u_{ij}/2)x_j x_i$, and by taking $a_{ij} = a_{ji} = u_{ij}/2$. The resulting matrix A is symmetric. We call it the *symmetric coefficient matrix* of the form.

Every quadratic form in n variables x_i can be written as $\mathbf{x}^T A \mathbf{x}$, where $\mathbf{x}$ is the column vector of variables and A is a symmetric matrix.

Diagonalization of Quadratic Forms

We saw in Section 6.3 that a symmetric matrix can be diagonalized by an orthogonal matrix. That is, if A is an $n \times n$ symmetric matrix, there exists an $n \times n$ orthogonal change-of-coordinates matrix C such that $C^{-1}AC = D$, where D is diagonal. Recall that the diagonal entries of D are $\lambda_1, \lambda_2, \dots, \lambda_n$, where the λ_j are the (not necessarily distinct) eigenvalues of the matrix A , the j th column of C is a unit eigenvector corresponding to λ_j , and the column vectors of C form an orthonormal basis for $\mathbb{R}^n$.

Consider now a quadratic form $\mathbf{x}^T A \mathbf{x}$, where A is symmetric, and let C be an orthogonal diagonalizing matrix for A . Because $C^{-1} = C^T$, the substitution

$$\mathbf{x} = C\mathbf{t} \quad \text{Diagonalizing substitution}$$

changes our quadratic form as follows:

$$\begin{aligned} \mathbf{x}^T A \mathbf{x} &= (C\mathbf{t})^T A (C\mathbf{t}) = \mathbf{t}^T C^T A C \mathbf{t} = \mathbf{t}^T C^{-1} A C \mathbf{t} \\ &= \mathbf{t}^T D \mathbf{t} = \lambda_1 t_1^2 + \lambda_2 t_2^2 + \cdots + \lambda_n t_n^2, \end{aligned}$$

where the λ_j are the eigenvalues of A . We have thus diagonalized the quadratic form. The value of $\mathbf{x}^T A \mathbf{x}$ for any $\mathbf{x}$ in $\mathbb{R}^n$ is the same as the value of $\mathbf{t}^T D \mathbf{t}$ for $\mathbf{t} = C^{-1}\mathbf{x}$.

EXAMPLE 4 Find a substitution $\mathbf{x} = C\mathbf{t}$ that diagonalizes the form $3x_1^2 + 10x_1x_2 + 3x_2^2$, and find the corresponding diagonalized form.

SOLUTION The symmetric coefficient matrix for the quadratic form is

$$A = \begin{bmatrix} 3 & 5 \\ 5 & 3 \end{bmatrix}.$$

We need to find the eigenvalues and eigenvectors for A . We have

$$|A - \lambda I| = \begin{vmatrix} 3 - \lambda & 5 \\ 5 & 3 - \lambda \end{vmatrix} = \lambda^2 - 6\lambda - 16 = (\lambda + 2)(\lambda - 8).$$

Thus, $\lambda_1 = -2$ and $\lambda_2 = 8$ are eigenvalues of A . Finding the eigenvectors that are to become the columns of the substitution matrix C , we compute

$$A - \lambda_1 I = A + 2I = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

and

$$A - \lambda_2 I = A - 8I = \begin{bmatrix} -5 & 5 \\ 5 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}.$$

Thus eigenvectors are $\mathbf{v}_1 = [-1, 1]$ and $\mathbf{v}_2 = [1, 1]$. Normalizing them to length 1 and placing them in the columns of the substitution matrix C , we obtain the diagonalizing substitution

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = C \begin{bmatrix} t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}.$$

Our theory then tells us that making the variable substitution

$$\begin{aligned} x_1 &= \left(\frac{1}{\sqrt{2}} \right) (-t_1 + t_2), \\ x_2 &= \left(\frac{1}{\sqrt{2}} \right) (t_1 + t_2), \end{aligned} \tag{5}$$

in the form $3x_1^2 + 10x_1x_2 + 3x_2^2$ will give the diagonal form

$$\lambda_1 t_1^2 + \lambda_2 t_2^2 = -2t_1^2 + 8t_2^2.$$

This can, of course, be checked by actually substituting the expressions for x_1 and x_2 in Eqs. (5) into the form $3x_1^2 + 10x_1x_2 + 3x_2^2$. ■

In the preceding example, we might like to clear denominators in our substitution given in Eqs. (5) by using $\mathbf{x} = (\sqrt{2}C)\mathbf{t}$, so that $x_1 = -t_1 + t_2$ and $x_2 = t_1 + t_2$. Using the substitution $\mathbf{x} = kC\mathbf{t}$ with a quadratic form $\mathbf{x}^T A \mathbf{x}$, we obtain

$$\mathbf{x}^T A \mathbf{x} = (kC\mathbf{t})^T A (kC\mathbf{t}) = k^2(\mathbf{t}^T C^T A C \mathbf{t}) = k^2(\mathbf{t}^T D \mathbf{t}).$$

Thus, with $k = \sqrt{2}$, the substitution $x_1 = -t_1 + t_2$, $x_2 = t_1 + t_2$ in Example 1 results in the diagonal form

$$2(-2t_1^2 + 8t_2^2) = -4t_1^2 + 16t_2^2.$$

In any particular situation, we would have to balance the desire for arithmetic simplicity against the desire to have the new orthogonal basis actually be orthonormal.

As an orthogonal matrix, C has determinant ± 1 . (See Exercise 22 in Section 6.3.) We state without proof the significance of the sign of $\det(C)$. If $\det(C) = 1$, then the new ordered orthonormal basis B given by the column vectors of C has the **same orientation** in $\mathbb{R}^n$ as the standard ordered basis E ; while if $\det(C) = -1$, then B has the **opposite orientation** to E . In order for $B = (\mathbf{b}_1, \mathbf{b}_2)$ in $\mathbb{R}^2$ to have the same orientation as $(\mathbf{e}_1, \mathbf{e}_2)$, there must be a *rotation* of the plane, given by a matrix transformation $\mathbf{x} = C\mathbf{t}$, which carries both $\mathbf{e}_1$ to $\mathbf{b}_1$ and $\mathbf{e}_2$ to $\mathbf{b}_2$. The same interpretation in terms of rotation is true in $\mathbb{R}^3$, with the

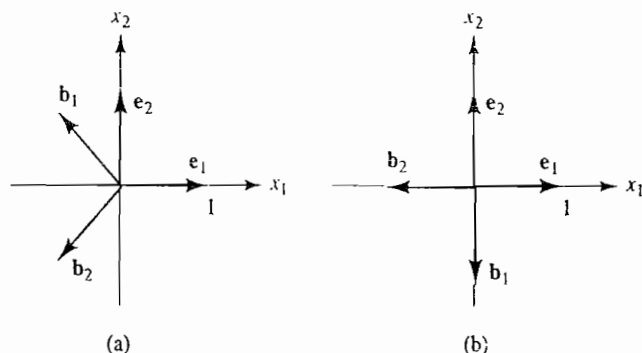


FIGURE 8.1

- (a) Rotation of axes, hence (b_1, b_2) and (e_1, e_2) have the same orientation;
 (b) not a rotation of axes, hence, (b_1, b_2) and (e_1, e_2) have opposite orientations.

additional condition that e_3 be carried to b_3 . To illustrate for the plane $\mathbb{R}^2$, Figure 8.1(a) shows an ordered orthonormal basis $B = (b_1, b_2)$, where $b_1 = [-1/\sqrt{2}, 1/\sqrt{2}]$ and $b_2 = [-1/\sqrt{2}, -1/\sqrt{2}]$, having the same orientation as E . Notice that

$$\det(C) = \begin{vmatrix} -1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{vmatrix} = \frac{1}{2} + \frac{1}{2} = 1.$$

$b_1 \qquad b_2$

Counterclockwise rotation of the plane through an angle of 135° carries E into B , preserving the order of the vectors. However, we see that the basis $(b_1, b_2) = ([0, -1], [-1, 0])$ shown in Figure 8.1(b) does not have the same orientation as E , and this time

$$\det(C) = \begin{vmatrix} 0 & -1 \\ -1 & 0 \end{vmatrix} = -1.$$

$b_1 \qquad b_2$

For any orthogonal matrix C having determinant -1 , multiplication of any *single* column of C by -1 gives an orthogonal matrix with determinant 1 . For the diagonalization we are considering, where the columns are normalized eigenvectors, multiplication by -1 still gives a normalized eigenvector. Although there will be some sign changes in the diagonalizing substitution, the final diagonal form remains the same, because the eigenvalues are the same and their order has not been changed.

We summarize all our work in one main theorem, and then conclude with a final example. Section 8.2 will give an application of the theorem to geometry, and Section 8.3 will give an application to optimization.

THEOREM 8.1 Principal Axis Theorem

Every quadratic form $f(\mathbf{x})$ in n variables $x_1, x_2, \dots, x_n$ can be diagonalized by a substitution $\mathbf{x} = C\mathbf{t}$, where C is an $n \times n$ orthogonal matrix. The diagonalized form appears as

$$\lambda_1 t_1^2 + \lambda_2 t_2^2 + \cdots + \lambda_n t_n^2,$$

where the λ_j are the eigenvalues of the symmetric coefficient matrix A of $f(\mathbf{x})$. The j th column vector of C is a normalized eigenvector $\mathbf{v}_j$ of A corresponding to λ_j . Moreover, C can be chosen so that $\det(C) = 1$.

We box a step-by-step outline for diagonalizing a quadratic form.

Diagonalizing a Quadratic Form $f(\mathbf{x})$

- Step 1** Find the symmetric coefficient matrix A of the form $f(\mathbf{x})$.
- Step 2** Find the (not necessarily distinct) eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of A .
- Step 3** Find an orthonormal basis for $\mathbb{R}^n$ consisting of normalized eigenvectors of A .
- Step 4** Form the matrix C , whose columns are the basis vectors found in step 3, in the order corresponding to the listing of eigenvalues in step 2. The transformation $\mathbf{x} = C\mathbf{t}$ is a **rotation** if $\det(C) = 1$. If a rotation is desired and $\det(C) = -1$, change the sign of all components of just one column vector in C .
- Step 5** The substitution $\mathbf{x} = C\mathbf{t}$ transforms $f(\mathbf{x})$ to the diagonal form $\lambda_1 t_1^2 + \lambda_2 t_2^2 + \cdots + \lambda_n t_n^2$.

EXAMPLE 5 Find a variable substitution that diagonalizes the form $2xy + 2xz$, and give the resulting diagonal form.

SOLUTION The symmetric coefficient matrix for the given quadratic form is

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Expanding the determinant on the last column, we find that

$$\begin{aligned} |A - \lambda I| &= \begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 0 \\ 1 & 0 & -\lambda \end{vmatrix} = 1(\lambda) - \lambda(\lambda^2 - 1) \\ &= -\lambda(-1 + \lambda^2 - 1) = -\lambda(\lambda^2 - 2). \end{aligned}$$

The eigenvalues of A are then $\lambda_1 = 0$, $\lambda_2 = \sqrt{2}$, and $\lambda_3 = -\sqrt{2}$.

Computing eigenvectors, we have

$$A - \lambda_1 I = A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix},$$

so $\mathbf{v}_1 = [0, -1, 1]$ is an eigenvector for $\lambda_1 = 0$. Moreover,

$$\begin{aligned} A - \lambda_2 I = A - \sqrt{2}I &= \begin{bmatrix} -\sqrt{2} & 1 & 1 \\ 1 & -\sqrt{2} & 0 \\ 1 & 0 & -\sqrt{2} \end{bmatrix} \sim \begin{bmatrix} 1 & -\sqrt{2} & 0 \\ -\sqrt{2} & 1 & 1 \\ 1 & 0 & -\sqrt{2} \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & -\sqrt{2} & 0 \\ 0 & -1 & 1 \\ 0 & \sqrt{2} & -\sqrt{2} \end{bmatrix} \sim \begin{bmatrix} 1 & -\sqrt{2} & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}, \end{aligned}$$

so $\mathbf{v}_2 = [\sqrt{2}, 1, 1]$ is an eigenvector for $\lambda_2 = \sqrt{2}$. In a similar fashion, we find that $\mathbf{v}_3 = [-\sqrt{2}, 1, 1]$ is an eigenvector for $\lambda_3 = -\sqrt{2}$. Thus,

$$C = \begin{bmatrix} 0 & \sqrt{2}/2 & -\sqrt{2}/2 \\ -1/\sqrt{2} & \frac{1}{2} & \frac{1}{2} \\ 1/\sqrt{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

is an orthogonal diagonalizing matrix for A . The substitution

$$\begin{aligned} x &= \left(\frac{1}{\sqrt{2}}\right)(t_2 - t_3) \\ y &= \frac{1}{2}(-\sqrt{2}t_1 + t_2 + t_3) \\ z &= \frac{1}{2}(\sqrt{2}t_1 + t_2 + t_3) \end{aligned}$$

will diagonalize $2xy + 2xz$ to become $\sqrt{2}t_2^2 - \sqrt{2}t_3^2$. ■

SUMMARY

Let $\mathbf{x}$ and $\mathbf{t}$ be $n \times 1$ column vectors with entries x_i and t_i , respectively.

1. For every nonzero $n \times n$ matrix A , the product $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ gives a quadratic form in the variables x_i .
2. Given any quadratic form $f(\mathbf{x})$ in the variables x_i , there exist an upper-triangular coefficient matrix U such that $f(\mathbf{x}) = \mathbf{x}^T U \mathbf{x}$ and symmetric coefficient matrix A such that $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$.
3. Any quadratic form $f(\mathbf{x})$ can be orthogonally diagonalized by using a variable substitution $\mathbf{x} = C\mathbf{t}$ as described in the boxed steps preceding Example 5.

EXERCISES

In Exercises 1–8, find the upper-triangular coefficient matrix U and the symmetric coefficient matrix A of the given quadratic form.

1. $3x^2 - 6xy + y^2$ 2. $8x^2 + 9xy - 3y^2$

3. $x^2 - y^2 - 4xy + 3xz - 8yz$

4. $x_1^2 - 2x_2^2 + x_3^2 + 6x_4^2 - 2x_1x_4 + 6x_2x_4 - 8x_1x_3$

5. $[x, y] \begin{bmatrix} -2 & 1 \\ 7 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

6. $[x, y] \begin{bmatrix} 7 & -10 \\ 15 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

7. $[x, y, z] \begin{bmatrix} 8 & 3 & 1 \\ 2 & 1 & -4 \\ -5 & 2 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

8. $[x_1, x_2, x_3, x_4] \begin{bmatrix} 2 & -1 & 3 & 0 \\ 4 & 2 & -1 & 3 \\ -6 & 3 & 0 & 7 \\ 10 & 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$

In Exercises 9–16, find an orthogonal substitution that diagonalizes the given quadratic form, and find the diagonalized form.

9. $2xy$

10. $3x^2 + 4xy$

11. $-6xy + 8y^2$

12. $x^2 + 2xy + y^2$

13. $3x^2 - 4xy + 3y^2$

14. $x_2^2 + 2x_1x_3$

15. $x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 - 2x_2x_3$
[SUGGESTION: Use Example 4 in Section 6.3.]

16. $x_1^2 + 2x_2^2 - 6x_1x_3 + x_3^2 - 4x_4^2$

17. Find a necessary and sufficient condition on a , b , and c such that the quadratic form $ax^2 + 2bxy + cy^2$ can be orthogonally diagonalized to kt^2 .

18. Repeat Exercise 17, but require also that $k = 1$.

 In Exercises 19–24, use the routine MATCOMP in LINTEK, or MATLAB, to find a diagonal form into which the given form can be transformed by an orthogonal substitution. Do not give the substitution.

19. $3x^2 + 4xy - 5y^2$

20. $x^2 - 8xy + y^2$

21. $3x^2 + y^2 - 2z^2 - 4xy + 6yz$

22. $y^2 - 8z^2 + 3xy - 4xz + 7yz$

23. $x_1^2 - 3x_1x_4 + 5x_4^2 - 8x_2x_3$

24. $x_1^2 - 8x_1x_2 + 6x_2x_3 - 4x_3x_4$

8.2

APPLICATIONS TO GEOMETRY

Conic Sections in $\mathbb{R}^2$

Figure 8.2 shows three different types of plane curves obtained when a double right-circular cone is cut by a plane. These *conic sections* are the ellipse, the hyperbola, and the parabola. Figure 8.3(a) shows an ellipse in *standard position*, with center at the origin. The equation of this ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \quad (1)$$

The ellipse in Figure 8.3(b) with center (h, k) has the equation

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1. \quad (2)$$

If we make the translation substitution $\bar{x} = x - h$, $\bar{y} = y - k$, then Eq. (2) becomes $\bar{x}^2/a^2 + \bar{y}^2/b^2 = 1$, which resembles Eq. (1).

By completing the square, we can put any quadratic polynomial equation in x and y with no xy -term but with x^2 and y^2 having coefficients of the same sign into the form of Eq. (2), possibly with 0 or -1 on the right-hand side. The procedure should be familiar to you from work with circles. We give one illustration for an ellipse.

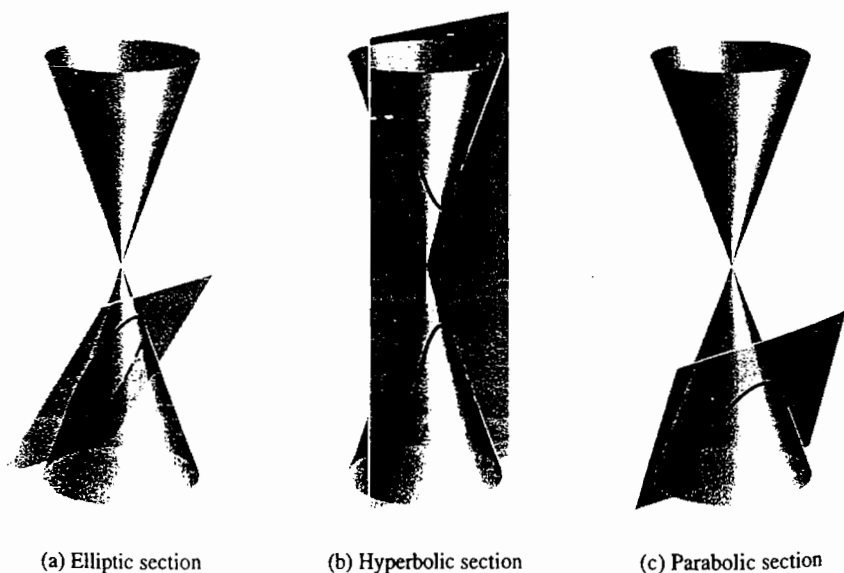


FIGURE 8.2

Sections of a cone: (a) an elliptic section; (b) a hyperbolic section; (c) a parabolic section.

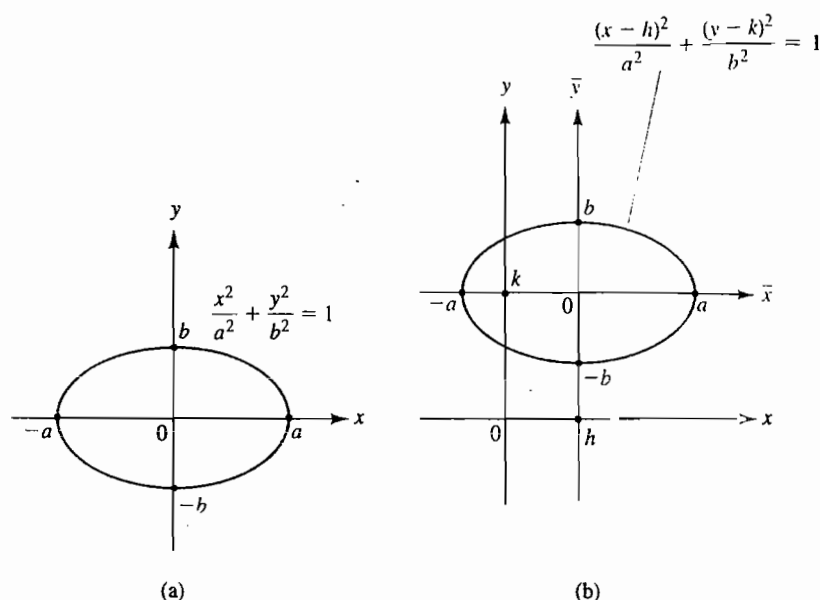


FIGURE 8.3

(a) Ellipse in standard position; (b) ellipse centered at (h, k) .**EXAMPLE 1** Complete the square in the equation

$$x^2 + 3y^2 - 4x + 6y = -1.$$

SOLUTION Completing the square, we obtain

$$\begin{aligned}(x^2 - 4x) + 3(y^2 + 2y) &= -1 \\(x - 2)^2 + 3(y + 1)^2 &= 4 + 3 - 1 = 6 \\ \frac{(x - 2)^2}{6} + \frac{(y + 1)^2}{2} &= 1,\end{aligned}$$

HISTORICAL NOTE ANALYTIC GEOMETRY is generally considered to have been founded by René Descartes (1596–1650) and Pierre Fermat (1601–1665) in the first half of the seventeenth century. But it was not until the appearance of a Latin version of Descartes' *Geometry* in 1661 by Frans van Schooten (1615–1660) that its influence began to be felt. This Latin version was published along with many commentaries; in particular, the *Elements of Curves* by Jan de Witt (1625–1672) gave a systematic treatment of conic sections. De Witt gave canonical forms of these equations similar to those in use today; for example, $y^2 = ax$, $by^2 + x^2 = f^2$, and $x^2 - by^2 = f^2$ represented the parabola, ellipse, and hyperbola, respectively. He then showed how, given an arbitrary second-degree equation in x and y , to find a transformation of axes that reduces the given equation to one of the canonical forms. This is, of course, equivalent to diagonalizing a particular symmetric matrix.

De Witt was a talented mathematician who, because of his family background, could devote but little time to mathematics. In 1653 he became in effect the Prime Minister of the Netherlands. Over the next decades, he guided the fortunes of the country through a most difficult period, including three wars with England. In 1672 the hostility of one of the Dutch factions culminated in his murder.

which is of the form of Eq. (2). This is the equation of an ellipse with center $(h, k) = (2, -1)$. Setting $\bar{x} = x - 2$ and $\bar{y} = y + 1$, we obtain the equation

$$\frac{\bar{x}^2}{6} + \frac{\bar{y}^2}{2} = 1.$$

If the constant on the right-hand side of the initial equation in Example 1 had been -7 , we would have obtained $\bar{x}^2 + 3\bar{y}^2 = 0$, which describes the single point $(\bar{x}, \bar{y}) = (0, 0)$. This single point is regarded as a *degenerate* ellipse. If the constant had been -8 , we would have obtained $\bar{x}^2 + 3\bar{y}^2 = -1$, which has no real solution; the ellipse is then called *empty*.

Thus, every polynomial equation

$$c_1x^2 + c_2y^2 + c_3x + c_4y = d, \quad c_1c_2 > 0, \quad (3)$$

describes an ellipse, possibly degenerate or empty.

In a similar fashion, an equation

$$c_1x^2 + c_2y^2 + c_3x + c_4y = d, \quad c_1c_2 < 0, \quad (4)$$

describes a hyperbola. Notice that the coefficients of x^2 and y^2 have *opposite* signs. The standard form of the equation for a hyperbola centered at $(0, 0)$ is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{or} \quad -\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

as shown in Figure 8.4. The dashed diagonal lines $y = \pm(b/a)x$ shown in the figure are the *asymptotes* of the hyperbola. By completing squares and

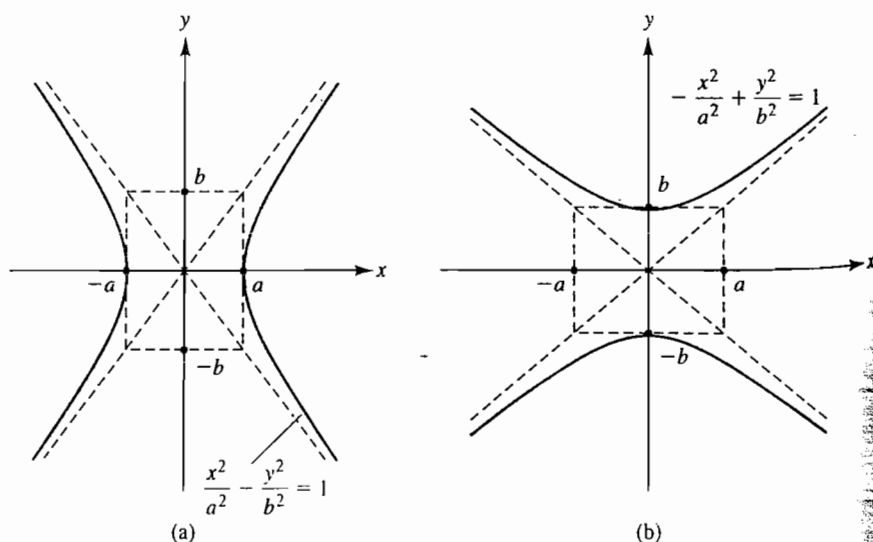


FIGURE 8.4

(a) The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$; (b) the hyperbola $-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

translating axes, we can reduce Eq. (4) to one of the standard forms shown in Figure 8.4, in variables $\bar{x}$ and $\bar{y}$, unless the constant in the final equation reduces to zero. In that case we obtain

$$\frac{\bar{x}^2}{a^2} - \frac{\bar{y}^2}{b^2} = 0 \quad \text{or} \quad \bar{y} = \pm \frac{b}{a} \bar{x}.$$

These equations represent two lines that can be considered a degenerate hyperbola. Thus any equation of the form of Eq. (4) describes a (possibly degenerate) hyperbola.

Finally, the equations

$$c_1x^2 + c_2x + c_3y = d \quad \text{and} \quad c_1y^2 + c_2x + c_3y = d, \quad c_1 \neq 0, \quad (5)$$

describe parabolas. If $c_3 \neq 0$ in the first equation in Eqs. (5) and $c_2 \neq 0$ in the second, these equations can be reduced to the form

$$\bar{x}^2 = a\bar{y} \quad \text{and} \quad \bar{y}^2 = a\bar{x} \quad (6)$$

by completing the square and translating axes. Figure 8.5 shows two parabolas in standard position. If Eqs. (5) reduce to $c_1x^2 + c_2x = d$ and $c_1y^2 + c_3y = d$, each describes two parallel lines that can be considered degenerate parabolas.

In summary, every equation of the form

$$c_1x^2 + c_2y^2 + c_3x + c_4y = d \quad (7)$$

with at least one of c_1 or c_2 nonzero describes a (possibly degenerate or empty) ellipse, hyperbola, or parabola.

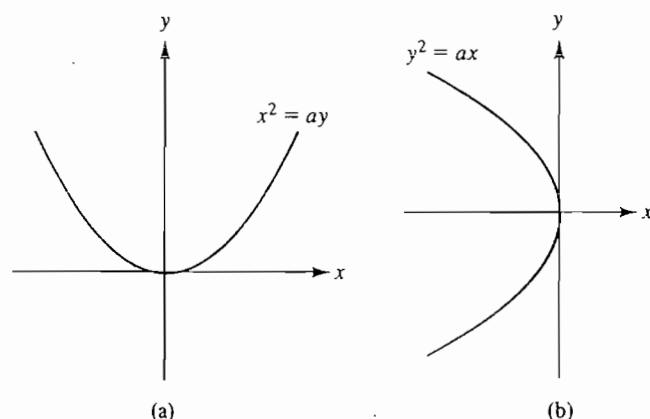


FIGURE 8.5
(a) The parabola $x^2 = ay$, $a > 0$; (b) the parabola $y^2 = ax$, $a < 0$.

Classification of Second-Degree Curves

We can now apply our work in Section 8.1 on diagonalizing quadratic forms to classification of the plane curves described by an equation of the type

$$ax^2 + bxy + cy^2 + dx + ey + f = 0 \quad \text{for } a, b, c \text{ not all zero.} \quad (8)$$

We make a substitution

$$\begin{bmatrix} x \\ y \end{bmatrix} = C \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}, \quad \text{where } \det(C) = 1, \quad (9)$$

which orthogonally diagonalizes the quadratic-form part of Eq. (8), which is in color, and we obtain an equation of the form

$$\lambda_1 t_1^2 + \lambda_2 t_2^2 + g t_1 + h t_2 + k = 0. \quad (10)$$

This equation has the form of Eq. (7) and describes an ellipse, hyperbola, or parabola. Remember that Eq. (9) corresponds to a rotation that carries the vector $\mathbf{e}_1$ to the first column vector $\mathbf{b}_1$ of C and carries $\mathbf{e}_2$ to the second column vector $\mathbf{b}_2$. We think of $\mathbf{b}_1$ as pointing out the t_1 -axis and $\mathbf{b}_2$ as pointing out the t_2 -axis. We summarize our work in a theorem and give just one illustration, leaving others to the exercises.

THEOREM 8.2 Classification of Second-Degree Plane Curves

Every equation of the form of Eq. (8) can be reduced to an equation of the form of Eq. (10) by means of an orthogonal substitution corresponding to a rotation of the plane. The coefficients λ_1 and λ_2 in Eq. (10) are the eigenvalues of the symmetric coefficient matrix of the quadratic-form portion of Eq. (8). The curve describes a (possibly degenerate or empty)

$$\begin{array}{ll} \text{ellipse} & \text{if } \lambda_1 \lambda_2 > 0, \\ \text{hyperbola} & \text{if } \lambda_1 \lambda_2 < 0, \\ \text{parabola} & \text{if } \lambda_1 \lambda_2 = 0. \end{array}$$

EXAMPLE 2 Use rotation and translation of axes to sketch the curve $2xy + 2\sqrt{2}x = 1$.

SOLUTION The symmetric coefficient matrix of the quadratic form $2xy$ is

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

We easily find that the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = -1$, and that

$$C = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

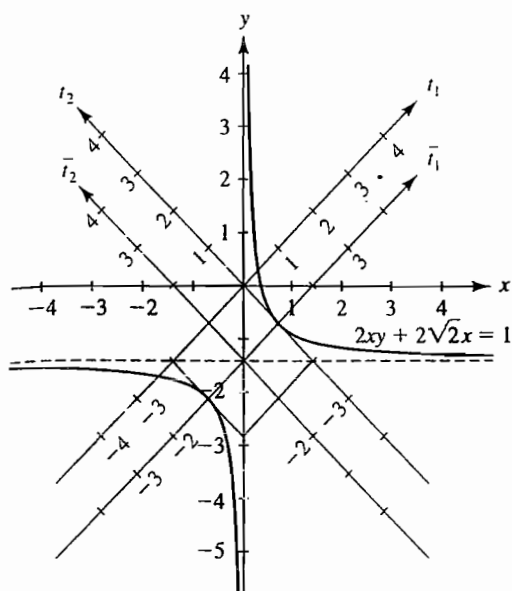


FIGURE 8.6
The hyperbola $2xy + 2\sqrt{2}x = 1$.

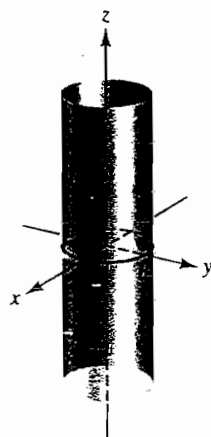


FIGURE 8.7
The elliptic cylinder $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

is an orthogonal diagonalizing matrix with determinant 1. The substitution

$$\begin{aligned}x &= \frac{1}{\sqrt{2}}(t_1 - t_2) \\ y &= \frac{1}{\sqrt{2}}(t_1 + t_2)\end{aligned}$$

then yields

$$t_1^2 - t_2^2 + 2t_1 - 2t_2 = 1.$$

Completing the square, we obtain

$$(t_1 + 1)^2 - (t_2 + 1)^2 = 1,$$

which describes the hyperbola shown in Figure 8.6. ■

Quadric Surfaces

An equation in three variables of the form

$$c_1x^2 + c_2y^2 + c_3z^2 + c_4x + c_5y + c_6z = d, \quad (11)$$

where at least one of c_1 , c_2 , or c_3 is nonzero, describes a *quadric surface* in space, which again might be degenerate or empty. Figures 8.7 through 8.15 show some of the quadric surfaces in standard position.

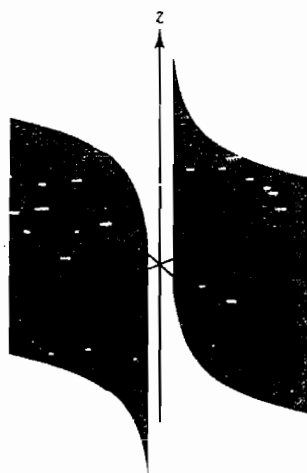


FIGURE 8.8
The hyperbolic cylinder
 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

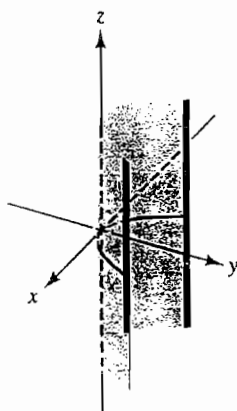


FIGURE 8.9
The parabolic cylinder
 $ay = x^2$.

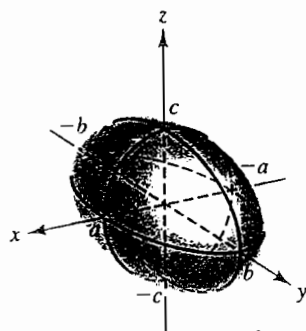


FIGURE 8.10
The ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

By completing the square in Eq. (11), which corresponds to translating axes, we see that Eq. (11) can be reduced to an equation involving $\bar{x}$, $\bar{y}$, and $\bar{z}$ in which a variable appearing to the second power does not appear to the first power. Notice that this is true for all the equations in Figures 8.7 through 8.15.

HISTORICAL NOTE THE EARLIEST CLASSIFICATION OF QUADRIC SURFACES was given by Leonhard Euler, in his precalculus text *Introduction to Infinitesimal Analysis* (1748). Euler's classification was similar to the conic-section classification of De Witt. Euler considered the second-degree equation in three variables $Ap^2 + Bq^2 + Cr^2 + Dpq + Epr + Fqr + Gp + Hq + Ir + K = 0$ as representing a surface in 3-space. As did De Witt, he gave canonical forms for these surfaces and showed how to rotate and translate the axes to reduce any given equation to a standard form such as $Ap^2 + Bq^2 + Cr^2 + K = 0$. An analysis of the signs of the new coefficients determined whether the given equation represented an ellipsoid, a hyperboloid of one or two sheets, an elliptic or hyperbolic paraboloid, or one of the degenerate cases. Euler did not, however, make explicit use of eigenvalues. He gave a general formula for rotation of axes in 3-space as functions of certain angles and then showed how to choose the angles to make the coefficients D , E , and F all zero.

Euler was the most prolific mathematician of all time. His collected works fill over 70 large volumes. Euler was born in Switzerland, but spent his professional life in St. Petersburg and Berlin. His texts on precalculus, differential calculus, and integral calculus (1748, 1755, 1768) had immense influence and became the bases for such texts up to the present. He standardized much of our current notation, introducing the numbers e , π , and i , as well as giving our current definitions for the trigonometric functions. Even though he was blind for the last 17 years of his life, he continued to produce mathematical papers almost up to the day he died.

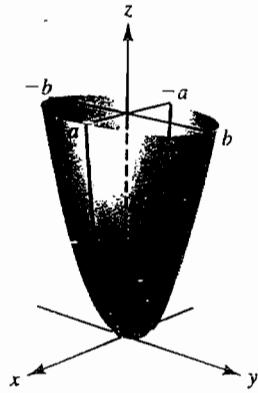


FIGURE 8.11

The elliptic paraboloid $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.

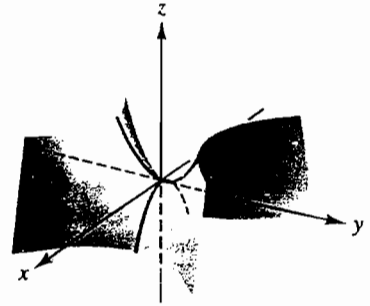


FIGURE 8.12

The hyperbolic paraboloid $z = \frac{y^2}{b^2} - \frac{x^2}{a^2}$.

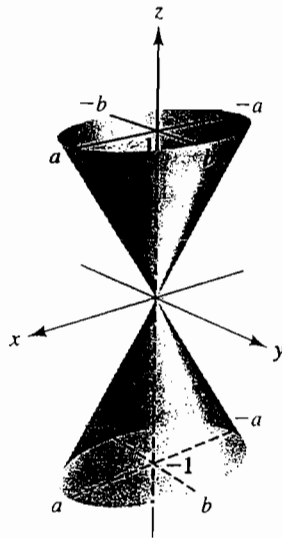


FIGURE 8.13

The elliptic cone $z^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.

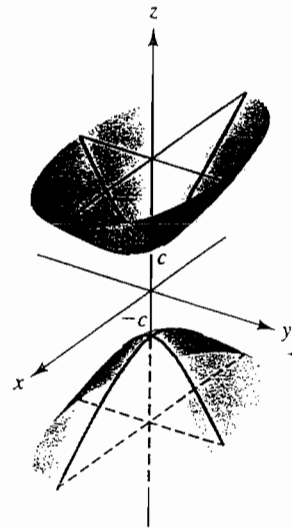


FIGURE 8.14

The hyperboloid of two sheets $\frac{z^2}{c^2} - 1 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.

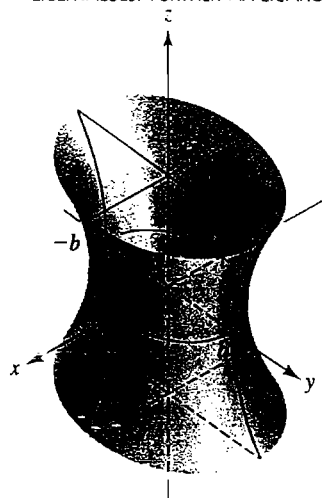


FIGURE 8.15

The hyperboloid of one sheet $\frac{z^2}{c^2} + 1 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.

Again, degenerate and empty cases are possible. For example, the equation $x^2 + 2y^2 + z^2 = -4$ gives an empty ellipsoid. The elliptic cone, hyperboloid of two sheets, and hyperboloid of one sheet in Figures 8.13 through 8.15 differ only in whether a constant in their equations is zero, negative, or positive.

Now consider a general second-degree polynomial equation

$$ax^2 + by^2 + cz^2 + dxy + exz + fyz + px + qy + rz + s = 0, \quad (12)$$

where the coefficient of at least one term of degree 2 is nonzero. Making a substitution

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = C \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix}, \quad \text{where } \det(C) = 1, \quad (13)$$

that orthogonally diagonalizes the quadratic-form portion of Eq. (12) (the portion in color), we obtain

$$\lambda_1 t_1^2 + \lambda_2 t_2^2 + \lambda_3 t_3^2 + p't_1 + q't_2 + r't_3 + s' = 0, \quad (14)$$

which is of the form of Eq. (11). We state this as a theorem.

THEOREM 8.3 Principal Axis Theorem for $\mathbb{R}^3$

Every equation of the form of Eq. (12) can be reduced to an equation of the form of Eq. (14) by an orthogonal substitution (13) that corresponds to a rotation of axes.